Group Assignment 1

Group 22

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1. Suppose that A and B are independent events, show that A^c and B^c are independent.

Solution 1:

$$\begin{aligned}
P(A^c \cap B^c) &= 1 - P(A \cup B)) \\
&= 1 - (P(A) + P(B) - P(A \cap B)) \\
&= 1 - (P(A) + P(B) - P(A)P(B)) \quad \text{(A \& B are independent)} \\
&= 1 - P(A) - P(B) + P(A)P(B) \\
&= (1 - P(A))(1 - P(B)) \\
&= P(A^c)P(B^c)
\end{aligned}$$

Therefore, A^c and B^c are independent.

2. The probability that a child has brown hair is 1/4. Assume independence between children and assume there are three children.
   (a) If it is known that at least one child has brown hair, what is the probability that at least two children have brown hair?
   (b) If it known that the oldest child has brown hair, what is the probability that at least two children have brown hair?

Solution 2(a):

Let:
Event A = ”at least two children have brown hair”
Event B = ”at least one child has brown hair”

$$P(\text{exactly 2 have brown hair}) = \binom{3}{2}\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^1 = 3 \cdot \frac{1}{16} \cdot \frac{3}{4} = \frac{9}{64}$$

$$P(\text{exactly 3 have brown hair}) = \binom{3}{3}\left(\frac{1}{4}\right)^3 = 1 \cdot \frac{1}{64} = \frac{1}{64}$$

$$P(A) = P(\text{exactly 2 have brown hair}) + P(\text{exactly 3 have brown hair}) = \frac{9}{64} + \frac{1}{64} = \frac{10}{64}$$

$$P(B) = 1 - P(\text{no child has brown hair}) = 1 - \left(\frac{3}{4}\right)^3 = \frac{37}{64}$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{10/64}{37/64} = \frac{10}{37}$$

Solution 2(b):

Let: Event C = "oldest child has brown hair"

$$
\begin{aligned}
P(A \mid C) &= 1 - P(\text{neither of the remaining 2 has brown hair}) \\
&= 1 - \left(\frac{3}{4}\right)^2 \\
&= 1 - \frac{9}{16} \\
&= \frac{7}{16}
\end{aligned}
$$

3. Let (X, Y) be uniformly distributed on the unit disc, $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$. Set $R = \sqrt{X^2 + Y^2}$. What is the CDF and PDF of R?

Solution 3:

Since (X, Y) are uniformly distributed on the unit disc, the CDF of R, which is defined as $R = \sqrt{X^2 + Y^2}$, is proportional to the area of the disc with radius r.

$$
\begin{aligned}
F_R(r) = P(R \leq r) &= \frac{\text{area of the disc with radius r}}{\text{area of the unit disc}} \\
&= \frac{\pi r^2}{\pi} \\
&= r^2, \quad \text{for some r} \in \text{[0,1]}
\end{aligned}
$$

Therefore, the CDF of R is

$$
F_R(r) = \begin{cases} 0, & r < 0 \\ r^2, & 0 \leq r \leq 1 \\ 1, & r > 1 \end{cases}
$$

To derive the PDF of R, we can take the derivative of CDF of R and it leads to

$$
f_R(r) = \begin{cases} 0, & r < 0 \\ 2r, & 0 \leq r \leq 1 \\ 0, & r > 1 \end{cases}
$$

4. A fair coin is tossed until a head appears. Let X be the number of tosses required. What is the expected value of X?

Solution 4:

The probability of landing heads on the first toss is 1/2. If heads is obtained on the first toss, then X=1. If tails is obtained on the first toss (probability 1/2), one toss has already been used, and the remaining expected number of tosses required to obtain heads for the first time remains E[X]. Hence, the equation holds. Therefore,

$$
\begin{aligned}
\mathbb{E}[X] &= \frac{1}{2} + \frac{1}{2}(1 + \mathbb{E}[X]) \\
\mathbb{E}[X] &= 1 + \frac{1}{2}\mathbb{E}[X] \\
\mathbb{E}[X] &= 2
\end{aligned}
$$

The expected value of X is 2.

5. Let $X_1, \ldots, X_n$ be IID from Bernoulli(p).

   (a) Let $\alpha > 0$ be fixed and define

   $$\varepsilon_n = \sqrt{\frac{1}{2n} \log \frac{2}{\alpha}}.$$

   Let $\hat{p}_n = \frac{1}{n}\sum_{i=1}^n X_i$ and define the confidence interval

   $$I_n = [\hat{p}_n - \varepsilon_n, \hat{p}_n + \varepsilon_n].$$

   Use Hoeffding's inequality to show that $\mathbb{P}(p \in I_n) \geq 1 - \alpha$.

   (b) Let $\alpha = 0.05$ and $p = 0.4$. Conduct a simulation study to see how often the confidence interval I_n contains p (called coverage). Do this for $n = 10, 100, 1000, 10000$. Plot the coverage as a function of n.

   (c) Plot the length of the confidence interval as a function of n.

   (d) Say that $X_1, \ldots, X_n$ represents if a person has a disease or not. Let us assume that unbeknownst to us the true proportion of people with the disease has changed from $p = 0.4$ to $p = 0.5$. We use the confidence interval to make a decision, that is when presented with evidence (samples) we calculate I_n and our decision is that the true proportion of people with the disease is in I_n. Conduct a simulation study to answer the following question: Given that the true proportion has changed, what is the probability that our decision is correct? Again using $n = 10, 100, 1000, 10000$.

Solution 5(a):

Hoeffding's inequality for IID Bernoulli (bounded in $[0,1]$) says

$$\mathbb{P}\left(|\hat{p}_n - p| \geq \varepsilon\right) \leq 2\exp(-2n\varepsilon^2).$$

Take $\varepsilon_n = \sqrt{\frac{1}{2n} \log \frac{2}{\alpha}}$. Then

$$2\exp(-2n\varepsilon_n^2) = 2\exp\left(-2n \cdot \frac{1}{2n} \log \frac{2}{\alpha}\right) = 2\exp\left(-\log \frac{2}{\alpha}\right) = \alpha.$$

So $\mathbb{P}(|\hat{p}_n - p| \geq \varepsilon_n) \leq \alpha$. Therefore

$$\mathbb{P}(p \in I_n) = \mathbb{P}(|\hat{p}_n - p| < \varepsilon_n) \geq 1 - \alpha.$$

That proves the claim.

Solution 5(b):

A simulation study was conducted to examine the coverage probability of the confidence interval $I_n = [\hat{p}_n - \varepsilon_n, \hat{p}_n + \varepsilon_n]$, where $\varepsilon_n = \sqrt{\frac{1}{2n} \log \frac{2}{\alpha}}$, for various sample sizes $n = 10, 100, 1000, 10000$. The true parameter was set to $p = 0.4$ and the significance level to $\alpha = 0.05$. For each sample size, 10,000 simulations were performed.

The results are shown in the figure below. As expected from Hoeffding's inequality, the coverage probability approaches the nominal level of $1 - \alpha = 0.95$ as the sample size n increases.

Solution 5(c):

The length of the confidence interval I_n is given by $2\varepsilon_n = 2\sqrt{\frac{1}{2n} \log \frac{2}{\alpha}}$. This length decreases as the sample size n increases, which is illustrated in the figure below.

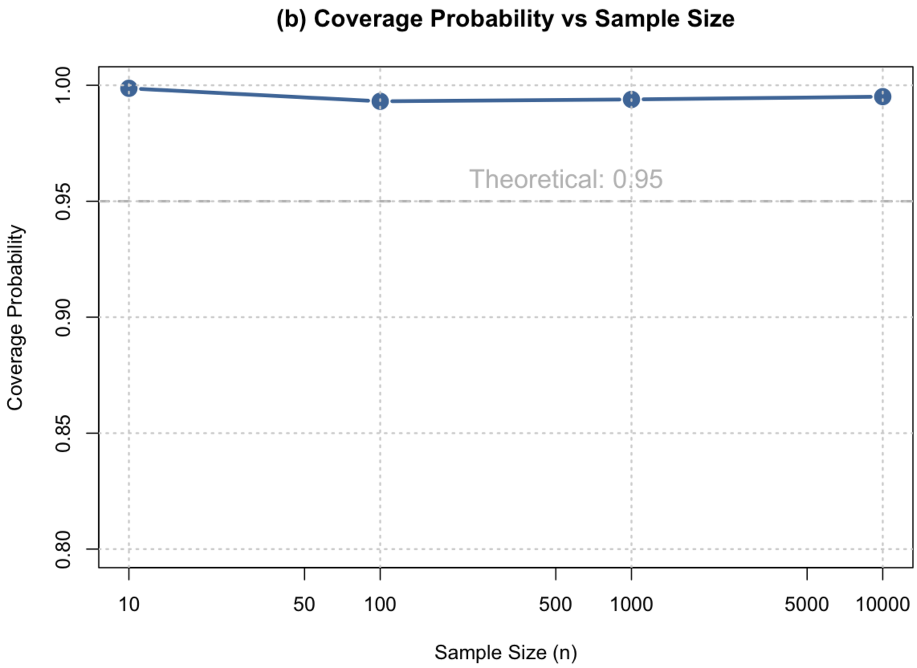

Figure 1: Coverage probability as a function of sample size n.

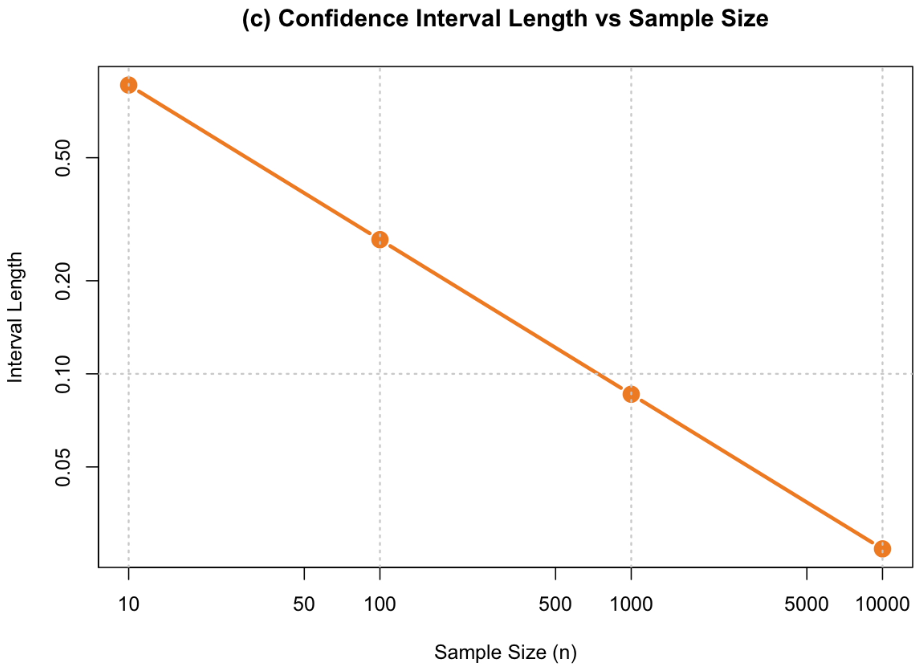

Figure 2: Length of the confidence interval as a function of sample size n.

Solution 5(d):

The probability that our decision is correct becomes higher when the sample size n increases. For small samples, like $n = 10$, this probability is very low (around 0.02). When $n = 100$, it increases to about 0.24, and when $n = 1000$, it is almost 1. This means that with larger samples, we can make the right decision much more often.

In simple words, small samples give very wide confidence intervals, so both $p = 0.4$ and $p = 0.5$

Table 1: Probability that our decision is correct for different sample sizes

Sample Size (n)	Probability that our decision is correct
10	0.0190
100	0.2370
1000	0.9998
10000	1.0000

can be inside the interval. As n gets larger, the interval becomes narrower and only includes the true value ($p = 0.5$). Therefore, our decision becomes almost always correct when the sample size is large enough.

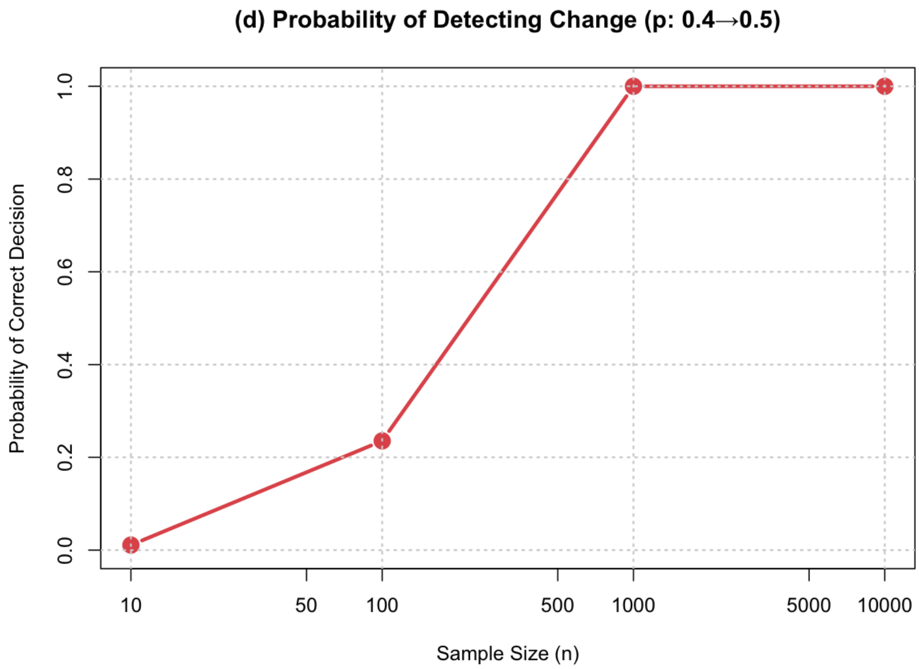

Figure 3: Probability of correct decision as a function of sample size n.